coursera

Depressive and Bipolar Disorders ✕

Today's goal • 3/3 items

Module 1
Learn With PsycLearn Essentials

Module 2
Opening to Depressive and Bipolar Disorders

Lesson 1: Culture and Mood Disorders
✓ Focus of the Course
Reading • 2 min
✓ Classifying Psychopathology
Reading • 4 min
✓ Culture and Psychopathology
Reading • 4 min
✓ Culture and Intersectionality
Reading • 4 min
✓ Culture and Stigma
Reading • 4 min
✓ Check Your Understanding: Culture and Mood Disorders
Practice Assignment • Grade: 100%

Lesson 2: Introduction to Depressive and Bipolar Disorders
✓ Depressive and Bipolar Disorders
Video • 6 min
✓ Living with a Mood Disorder: A Case Study
Reading • 10 min
✓ Overview of Depressive and Bipolar Disorders
Reading • 10 min
✓ Check Your Understanding: Introduction to Depressive and Bipolar Disorders
Practice Assignment • Grade: 80%

Key Takeaways
✓ Key Takeaways: Opening to Depressive and Bipolar Disorders
Reading • 10 min

Module 2 Quiz
✓ Module Quiz: Opening to Depressive and Bipolar Disorders
Graded Assignment • Grade: 87.5%

Module 3
Depressive and Bipolar Disorders

Lesson 3: Overview of Depressive Disorders
✓ Depression: A Case Study
Reading • 5 min
✓ What Does Depression Feel Like?
Reading • 7 min
✓ Diagnostic Criteria and Prevalence for Depressive Disorders
Reading • 5 min
○ Check Your Understanding: Overview of Depressive Disorders
Practice Assignment • 12 min

Lesson 4: Overview of Bipolar Disorders
✓ Bipolar: A Case Study
Reading • 5 min
✓ What Does Bipolar Feel Like?
Reading • 7 min
✓ Diagnostic Criteria and Prevalence for Bipolar Disorders
Reading • 5 min
✓ Check Your Understanding: Overview of Bipolar Disorders
Practice Assignment • Grade: 100%

Key Takeaways
✓ Key Takeaways: Depressive and Bipolar Disorders
Reading • 10 min

Module 3 Quiz
✓ Module Quiz: Depressive and Bipolar Disorders
Graded Assignment • Grade: 87.5%

Module 4
Biological Contributing Factors

Lesson 5: Genetic Contributing Factors
✓ Developmental Psychopathology
Reading • 3 min
✓ Genetic Contributing Factors
Reading • 3 min
✓ The Link Between Genes and Psychopathology
Video • 4 min
✓ Practical Scenario
Real-world application • 5 min
○ **Genetic Contributing Factors for Depression**
Reading • 10 min
○ Epigenetic Contributors to Depression
Video • 3 min
○ Genetic Contributing Factors for Bipolar Disorder
Reading • 3 min
○ Check Your Understanding: Genetic Contributing Factors
Practice Assignment • 4 min

Lesson 6: Neurobiological Contributing Factors
○ Depressive Disorders: A Case Study
Reading • 5 min
○ Communication in the Synapse
Video • 4 min
○ Neurochemical Contributors to Depression
Reading • 10 min
○ Practical Scenario
Real-world application • 5 min
○ Hormonal Contributors to Depression
Reading • 5 min
○ Brain Regions Linked to Depression
Video • 4 min
○ Neurobiological Contributors to Depressive and Bipolar Disorders
Reading • 15 min
○ Check Your Understanding: Biological Contributing Factors
Practice Assignment • 6 min

Key Takeaways
○ Key Takeaways: Biological Contributing Factors
Reading • 10 min

Module 4 Quiz
○ Module Quiz: Biological Contributing Factors
Graded Assignment • 9 min

Module 5
Psychological Contributing Factors

Module 6
Treatments for Depressive and Bipolar Disorders

Module 7
Suicide

Module 8
Conclusion

Module 9
Cumulative Quiz

Module 10
Course Resources

Module 11
PsycLearn Essentials APA Student Resources

Genetic Contributing Factors for Depression

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Genetic Contributing Factors for Depression

Epidemiologist Myrna Weissman and her colleagues examined the risk of developing depression in the children and the grandchildren of depressed parents (Weissman et al., 2016). Given that depressive disorders are known to run in families, children of depressed parents should be more likely to develop depression. Weissman and her colleagues found that the children of depressed parents were twice as likely as the children of nondepressed parents to develop depression. The familial influence was also borne out by the depression levels of the grandchildren. Children with a depressed parent and depressed grandparent had the highest risk for developing depression themselves (Goodman et al., 2011; Weissman et al., 2016).

A heritability rate of 30-50% for depression indicates that environmental factors also play a role in the development of depression (Sullivan et al., 2000). Studies typically find that shared environmental factors (environmental influences on both twins), such as family, household, and neighborhood variables have relatively little influence on depression. Rather, it is unique environmental influences (environmental influences on one twin but not the other) that are associated with the development of depression (Sullivan et al., 2000). Examples of unique environmental influences on depression include adverse childhood experiences and victimization, such as from bullying (Kwong et al, 2019).

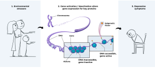
Epigenetics

There is evidence that gene expression is impacted by the environment. **Epigenetics** is the field of study that investigates the ways in which the environment can lead to the activation or deactivation of various parts of our genome. Environmental influences, such as stress, can influence the expression of genes, and increase or decrease the likelihood that a person will develop a physical or psychological disorder. Stress is a significant risk factor for depression. Maltreatment during childhood, for example, is an environmental factor that has been found to activate parts of the genome related to depression and suicidal ideation (Covington et al, 2009; McGowan et al., 2009).

The mechanism of epigenetics can be thought of as three basic steps (Park et al., 2019; Sun et al., 2013):

1. Environmental stressors make persistent changes to an individual's genome.
2. Genomic changes promote activation or deactivation of genes on the person's DNA, which promotes the expression or suppression of an important protein.
3. The protein influences neural and behavioral functions of the brain.

The Mechanism of Epigenetics


1. Environmental stressors
2. Gene activation / deactivation alters gene expression for key proteins
Chromosome
Epigenetic factor
DNA
Gene
DNA accessible, gene active
Histone
DNA inaccessible, gene inactive
3. Depressive symptoms

Research on identical twins has demonstrated that epigenetics plays a role in who manifests disorders such as depression, bipolar disorder, post-traumatic stress disorder, and schizophrenia (Bustamante & Uddin, 2014), which offers a explanation of how environmental influences can affect only one member of a twin pair.

Epigenetics has been used to explain the psychological impact across generations of major traumatic experiences such as the Holocaust and slavery (Lehrner & Yehuda, 2018). Adult children of fathers who were Holocaust survivors and developed posttraumatic stress disorder showed epigenetic markers on genes that have been suggested to be involved in posttraumatic stress (Yehuda et al., 2014). However, due to the logistical difficulties conducting well-controlled research in this area, evidence supporting the trans-generational transmission of trauma through epigenetics in humans is still not definitive (Yehuda & Lehrner, 2018) and more research is needed.

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